

A Practical Guide to Artificial neural network

TOPIC -- ARTIFICIAL NEURAL NETWORK

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

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Hardware Neural networks require enormous computing resources for training. While the brain requires only 20 watts of power, training commercial transformers requires (2026) data centers with hundreds of megawatts. From 1991 to 2015, computing power, especially as delivered by GPGPUs (on GPUs), increased around a million-fold, enabling the use of the backpropagation algorithm. Neuromorphic engineering or a physical neural network constructed non-von-Neumann chips to directly implement neural networks in circuitry. For example, Alphabet introduced custom chips (Tensor Processing Unit).

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Stochastic neural network Stochastic neural networks originating from Sherrington–Kirkpatrick models are a type of neural network built by introducing random variations into the network, either by giving neurons stochastic transfer functions, or by giving them stochastic weights. This makes them useful tools for optimization problems, since the random fluctuations help the network escape from local minima. Stochastic neural networks trained using a Bayesian approach are known as Bayesian neural networks.

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Lack of interpretability NNs are "black box" systems that have complicated understanding of their decision-making processes. NNs are further vulnerable to adversarial examples ("poison"), that can cause incorrect predictions. These concerns have led to increased research in explainable artificial intelligence (XAI), robust machine learning, and hybrid AI approaches that combine neural learning with symbolic reasoning. Advocates of hybrid models also say that such a mixture can better capture the mechanisms of the human mind. Progress has included local vs. non-local learning and shallow vs. deep architectures. Analyzing human thought (a biological neural network) has also proven difficult.

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The backpropagated value (secondary reinforcement) is the emotion toward the consequence situation. The CAA exists in two environments, its behavioral environment, and its genetic environment, from which it receives initial emotions (only once) about to be encountered situations in the behavioral environment. Having received the genome vector (species vector) from the genetic environment, the CAA will learn a goal-seeking behavior, in the behavioral environment that contains both desirable and undesirable situations.

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Error-based monitoring: comparing current predictions against ground-truth labels. This approach directly quantifies predictive performance but may be impractical when labels are delayed or impractical to obtain. Data distribution monitoring: detecting changes in the input data distribution using statistical tests, divergence measures, or density-ratio estimation. Representation monitoring: tracking the distribution of internal embeddings or hidden-layer features. Shifts in the latent representation can indicate nonstationarity even when labels are unavailable. Statistical methods such as statistical process control charts have been adapted for this purpose.

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Network design The choice of model depends on the data and the application. Models that work well with textual data are typically not the best choice for image data, etc. An important element is which training/learning the model uses. Neural architecture search (NAS) uses machine learning to automate NN design. NAS has yielded networks that compare well with hand-designed systems. The basic algorithm is to propose a candidate model, evaluate it against a dataset, and use the results as feedback to teach the NAS network. Efforts include AutoML and AutoKeras. scikit-learn library provides functions to help with building a deep network from scratch. Hyperparameters are design choices (they are not learned).

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Training/learning involves adjusting the weights of the network to improve the accuracy of the result. NNs typically require vast numbers of sample inputs (far more than biological brains) to achieve a given level of function. This is done by minimizing the observed errors among sample observations. Training takes place before a network is deployed, and (unlike brains) does not continue thereafter. Instead, the network may be retrained from scratch as more sample data becomes available. Empirical risk minimization adjusts node and link

weights to minimize the difference (empirical risk), between the predicted output and the known values in the training samples. A defined loss function measures the degree of error. Backpropagation spreads the error (adjusts the weights) from the output nodes across the network to the input nodes. The intent is to allow the network to process data that is not included in the training samples. As long as the value of the loss function (its cost) continues to decline, the network is continuing to improve. The function typically produces a statistic whose value is only approximate. When the cost is low, the difference between the output (e.g. almost certainly a cat) and the correct answer (a cat) is small. Most learning models can be viewed as a straightforward application of optimization theory and statistical estimation. Learning typically ends when additional observations do not usefully reduce the cost. The cost typically approaches, but does not reach, 0. If no feasible amount of sample data yields a low cost, training is labeled a failure.

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In machine learning, a neural network (NN) or neural net, is a computational model inspired by the structure and functions of biological neural networks. A neural network consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain. Artificial neuron models that mimic biological neurons more closely have also been recently investigated and shown to significantly improve performance. These are connected by edges, which model the synapses in the brain. Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons. The "signal" is a real number, and the output of each neuron is computed by some non-linear function of the totality of its inputs, called the activation function. The strength of the signal at each connection is determined by a weight, which adjusts as part of the training process. Groups of neurons are aggregated into layers. Each layer performs a transformation on its inputs. Signals travel from the first layer (the input layer) to the last layer (the output layer), typically passing through multiple intermediate layers (hidden layers). A network is typically called a deep neural network if it has at least two hidden layers. Deep neural networks are capable of learning sophisticated hierarchical representations. Training neural networks is a compute-intensive process, accelerated by the use of graphics processing units (GPUs), and large datasets. Architectural innovations such as convolutional neural networks (CNNs) significantly improved performance in computer vision tasks, while recurrent neural networks (RNNs) enabled modeling of sequential data such as speech and time-series information. Transformer architectures introduced attention mechanisms that allow neural networks to model long-range dependencies in data and have the basis of large language models. Artificial neural networks are

used for a myriad of tasks including chatbots, large-scale text, image, and video generation, and robotics.